

```
!pip install pyspark
```

Requirement already satisfied: pyspark in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: py4j<0.10.9.10,>=0.10.9.7 in /usr/local/lib/pythc

```
from pyspark.sql import SparkSession

spark = SparkSession.builder \
    .appName("RDD_Practice") \
    .master("local[*]") \
    .getOrCreate()
```

```
sc = spark.sparkContext
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

archive.zip

archive.zip(application/x-zip-compressed) - 45212625 bytes, last modified: 7/23/2026 - 100% done

Saving archive.zip to archive.zip

```
import zipfile
import os

# Assuming the uploaded file is named 'archive.zip'
zip_file_path = 'archive.zip'

# Check if the zip file exists
if os.path.exists(zip_file_path):
    with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
        zip_ref.extractall('.') # Extract all contents to the current directory
        print(f"Successfully unzipped '{zip_file_path}'")
        print("Extracted files:")
        for filename in zip_ref.namelist():
            print(f"- {filename}")
else:
    print(f"Error: Zip file '{zip_file_path}' not found.")
```

Successfully unzipped 'archive.zip'
Extracted files:
- faker_employee.csv

```
rdd = sc.textFile("/content/faker_employee.csv")
```

```
rdd.take(5)
```

```
['Employee ID,First Name,Last Name,Department,Salary,Joining Date,Email  
ID,Address',  
 '1,Nicholas,Martin,Sales,50819,2022-11-16,dannymays@example.com,"14632 Ashley  
Trafficway Suite 345',  
 'Lake Katrinaside, MH 96450"',  
 '2,Joy,Miller,Legal,58024,2023-02-03,heather59@example.net,"00670 Oliver  
Harbors',  
 'Port Andrewchester, TN 14223"']
```

```
header = rdd.first()
```

```
data = rdd.filter(lambda row: row != header)
```

```
employee_rdd = data.map(lambda row: row.split(","))
```

```
employee_rdd.take(3)
```

```
[[ '1',  
  'Nicholas',  
  'Martin',  
  'Sales',  
  '50819',  
  '2022-11-16',  
  'dannymays@example.com',  
  '"14632 Ashley Trafficway Suite 345',  
  'Lake Katrinaside', ' MH 96450"'],  
 [ '2',  
  'Joy',  
  'Miller',  
  'Legal',  
  '58024',  
  '2023-02-03',  
  'heather59@example.net',  
  '"00670 Oliver Harbors"']]
```

```
employee_rdd.count()
```

```
2000000
```

```
employee_rdd.first()
```

```
[ '1',  
  'Nicholas',  
  'Martin',  
  'Sales',  
  '50819',
```

```
'2022-11-16',  
'dannymays@example.com',  
'"14632 Ashley Trafficway Suite 345']
```

header

```
'Employee ID,First Name,Last Name,Department,Salary,Joining Date,Email ID,Address'
```

```
bangalore = employee_rdd.filter(lambda x: x[5] == "Bangalore")  
  
#bangalore.collect()
```

```
# Remove the header  
header = rdd.first()  
  
# Keep only data rows  
data_rdd = rdd.filter(lambda row: row != header)  
  
# Split each row into columns  
split_rdd = data_rdd.map(lambda row: row.split(","))  
  
# Filter Salary > 50000  
filtered_rdd = split_rdd.filter(lambda row: int(row[4]) > 50000)  
  
# Print  
for row in filtered_rdd.collect():  
    print(row)
```



```
-----
Py4JJavaError                                Traceback (most recent call last)
/tmp/ipykernel_1224/3302735668.py in <cell line: 0>()
    12
    13 # Print
--> 14 for row in filtered_rdd.collect():
    15     print(row)
```

3 frames

```
/usr/local/lib/python3.12/dist-packages/py4j/protocol.py in
get_return_value(answer, gateway_client, target_id, name)
    325         value = OUTPUT_CONVERTER[type](answer[2:], gateway_client)
    326         if answer[1] == REFERENCE_TYPE:
--> 327             raise Py4JJavaError(
    328                 "An error occurred while calling {0}{1}{2}.\n".
    329                 format(target_id, ".", name), value)
```

```
Py4JJavaError: An error occurred while calling
z:org.apache.spark.api.python.PythonRDD.collectAndServe.
: org.apache.spark.SparkException: Job aborted due to stage failure: Task 0 in
stage 18.0 failed 1 times, most recent failure: Lost task 0.0 in stage 18.0
(TID 36) (2a7142bddd77 executor driver):
org.apache.spark.api.python.PythonException: Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/worker.py", line 2044, in main
process()
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/worker.py", line 2036, in
process
    serializer.dump_stream(out_iter, outfile)
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/serializers.py", line 273, in
dump_stream
    vs = list(itertools.islice(iterator, batch))
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/util.py", line 131, in wrapper
    return f(*args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/tmp/ipykernel_1224/3302735668.py", line 11, in <lambda>
IndexError: list index out of range
```

```
at
org.apache.spark.api.python.BasePythonRunner$ReaderIterator.handlePythonExceptio
at
org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:940)
at
org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:925)
at
org.apache.spark.api.python.BasePythonRunner$ReaderIterator.hasNext(PythonRunner
at
org.apache.spark.InterruptibleIterator.hasNext(InterruptibleIterator.scala:37)
at scala.collection.mutable.Growable.addAll(Growable.scala:61)
at scala.collection.mutable.Growable.addAll$(Growable.scala:57)
at scala.collection.mutable.ArrayBuilder.addAll(ArrayBuilder.scala:75)
at scala.collection.IterableOnceOps.toArray(IterableOnce.scala:1505)
```

```

    at scala.collection.IterableOnceOps.toArray$(IterableOnce.scala:1498)
    at
org.apache.spark.InterruptibleIterator.toArray(InterruptibleIterator.scala:28)
    at org.apache.spark.rdd.RDD.$anonfun$collect$2(RDD.scala:1057)
    at org.apache.spark.SparkContext.$anonfun$runJob$5(SparkContext.scala:2524)
    at org.apache.spark.scheduler.ResultTask.runTask(ResultTask.scala:93)
    at org.apache.spark.TaskContext.runTaskWithListeners(TaskContext.scala:171)
    at org.apache.spark.scheduler.Task.run(Task.scala:147)
    at
org.apache.spark.executor.Executor$TaskRunner.$anonfun$run$5(Executor.scala:647)
    at
org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally(SparkErrorUtils.scala:8)
    at
org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally$(SparkErrorUtils.scala:
    at org.apache.spark.util.Utils$.tryWithSafeFinally(Utils.scala:99)
    at org.apache.spark.executor.Executor$TaskRunner.run(Executor.scala:650)
    at
java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.j
    at
java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.
    at java.base/java.lang.Thread.run(Thread.java:840)

```

Driver stacktrace:

```

    at
org.apache.spark.scheduler.DAGScheduler.$anonfun$abortStage$3(DAGScheduler.scala
    at scala.Option.getOrElse(Option.scala:201)
    at
org.apache.spark.scheduler.DAGScheduler.$anonfun$abortStage$2(DAGScheduler.scala
    at
org.apache.spark.scheduler.DAGScheduler.$anonfun$abortStage$2$adapted(DAGSchedul
    at scala.collection.immutable.List.foreach(List.scala:334)
    at
org.apache.spark.scheduler.DAGScheduler.abortStage(DAGScheduler.scala:2927)
    at
org.apache.spark.scheduler.DAGScheduler.$anonfun$handleTaskSetFailed$1(DAGSchedu
    at
org.apache.spark.scheduler.DAGScheduler.$anonfun$handleTaskSetFailed$1$adapted(D
    at scala.Option.foreach(Option.scala:437)
    at
org.apache.spark.scheduler.DAGScheduler.handleTaskSetFailed(DAGScheduler.scala:1
    at
org.apache.spark.scheduler.DAGSchedulerEventProcessLoop.doOnReceive(DAGScheduler
    at
org.apache.spark.scheduler.DAGSchedulerEventProcessLoop.onReceive(DAGScheduler.s
    at
org.apache.spark.scheduler.DAGSchedulerEventProcessLoop.onReceive(DAGScheduler.s
    at org.apache.spark.util.EventLoop$$anon$1.run(EventLoop.scala:50)
    at org.apache.spark.scheduler.DAGScheduler.runJob(DAGScheduler.scala:1009)
    at org.apache.spark.SparkContext.runJob(SparkContext.scala:2484)
    at org.apache.spark.SparkContext.runJob(SparkContext.scala:2505)
    at org.apache.spark.SparkContext.runJob(SparkContext.scala:2524)
    at org.apache.spark.SparkContext.runJob(SparkContext.scala:2549)
    at org.apache.spark.rdd.RDD.$anonfun$collect$1(RDD.scala:1057)
    at
org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
    at

```

```

org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
  at org.apache.spark.rdd.RDD.withScope(RDD.scala:417)
  at org.apache.spark.rdd.RDD.collect(RDD.scala:1056)
  at
org.apache.spark.api.python.PythonRDD$.collectAndServe(PythonRDD.scala:203)
  at org.apache.spark.api.python.PythonRDD.collectAndServe(PythonRDD.scala)
  at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke0(Native
Method)
  at
java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAcces
  at
java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMet
  at java.base/java.lang.reflect.Method.invoke(Method.java:569)
  at py4j.reflection.MethodInvoker.invoke(MethodInvoker.java:244)
  at py4j.reflection.ReflectionEngine.invoke(ReflectionEngine.java:374)
  at py4j.Gateway.invoke(Gateway.java:282)
  at py4j.commands.AbstractCommand.invokeMethod(AbstractCommand.java:132)
  at py4j.commands.CallCommand.execute(CallCommand.java:79)
  at
py4j.ClientServerConnection.waitForCommands(ClientServerConnection.java:184)
  at py4j.ClientServerConnection.run(ClientServerConnection.java:108)
  at java.base/java.lang.Thread.run(Thread.java:840)
Caused by: org.apache.spark.api.python.PythonException: Traceback (most recent
call last):
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/worker.py", line 2044, in main
process()
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/worker.py", line 2036, in
process
    serializer.dump_stream(out_iter, outfile)
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/serializers.py", line 273, in
dump_stream
Next steps = Explain the error
    tools.islice(iterator, batch))
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/local/lib/python3.12/dist-
packages/pyspark/python/lib/pyspark.zip/pyspark/util.py", line 131, in wrapper
    return f(*args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/tmp/ipykernel_1224/3302735668.py", line 11, in <lambda>
IndexError: list index out of range

  at
org.apache.spark.api.python.BasePythonRunner$ReaderIterator.handlePythonExceptio
  at
org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:940)
  at
org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:925)
  at
org.apache.spark.api.python.BasePythonRunner$ReaderIterator.hasNext(PythonRunner
  at
org.apache.spark.InterruptibleIterator.hasNext(InterruptibleIterator.scala:37)
  at scala.collection.mutable.Growable.addAll(Growable.scala:61)
  at scala.collection.mutable.Growable.addAll$(Growable.scala:57)
  at scala.collection.mutable.ArrayBuilder.addAll(ArrayBuilder.scala:75)
  at scala.collection.IterableOnceOps.toArray(IterableOnce.scala:1505)

```

```
print(rdd.take(10))
```

```
org.apache.spark.InterruptibleIterator.toArray(InterruptibleIterator.scala:28)
[Employee ID,First Name,Last Name,Department,salary,Joining Date,Email ID,Address]
at org.apache.spark.rdd.RDD.$anonfun$collect$2(RDD.scala:1057)
at org.apache.spark.SparkContext.$anonfun$runJob$5(SparkContext.scala:2524)
```

```
split_rdd = rdd.map(lambda x: x.split(","))
```

```
for row in split_rdd.take(10):
    print(row, len(row))
```

```
org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally(SparkErrorUtils.scala:8)
[Employee ID,First Name,Last Name,Department,salary,Joining Date,Email ID,Address]
[1,'at',Nicholas', 'Martin', 'Sales', '50819', '2022-11-16', 'dannymays@example.c
org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally$(SparkErrorUtils.scala:
[2,'at',969,apac,spark,Legal,Utils$8024,tryWithSafeFinally(Utils.scala:99)
[Port And,spark,executor$TaskRunner.run(Executor.scala:650)
[3,'at',Thomas', 'Roth', 'R&D', '54572', '2021-08-18', 'garyhogan@example.com', '
base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.j
North Willie', 'AR 99982']
[4,'at',Brittney', 'Morgan', 'Sales', '72174', '2022-10-03', 'wcervantes@example.
base/scala/scala/Thread2ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.
James', 'at', 'Katherine', 'Bell', 'Training and Development', '52848', '2022-08-16', 't
```

```
header = rdd.first()
```

```
filtered_rdd = (
    rdd
    .filter(lambda x: x != header)
    .map(lambda x: x.split(","))
    .filter(lambda row: len(row) >= 8)
    .filter(lambda row: row[4].strip().isdigit())
    .filter(lambda row: int(row[4]) > 70000)
)
```

```
for row in filtered_rdd.collect():
    print(row)
```



```
[
  '999884', 'Kelly', 'Cooper', 'Administration', '78866', '2021-02-23', 'mason@example.com',
  '999885', 'Ashley', 'Smith', 'Strategic Planning', '76164', '2022-03-11', 'jeff@example.com',
  '999889', 'Patrick', 'Watson', 'Corporate Communications', '75008', '2021-01-01', 'p.watson@example.com',
  '999891', 'Sabrina', 'Johnson', 'Business Development', '73914', '2021-08-06', 's.johnson@example.com',
  '999894', 'Steve', 'Phillips', 'Strategic Planning', '72341', '2023-06-18', 's.phillips@example.com',
  '999897', 'Timothy', 'Vargas', 'R&D', '75456', '2023-06-12', 'sspark@example.com',
  '999900', 'Jeffrey', 'Marshall', 'Compliance', '74860', '2022-11-08', 'jamie4@example.com',
  '999901', 'Michael', 'Fisher', 'Facilities Management', '79494', '2022-04-04', 'm.fisher@example.com',
  '999902', 'Lisa', 'Turner', 'R&D', '78843', '2023-01-26', 'donna51@example.com',
  '999905', 'Keith', 'Baker', 'PR', '73280', '2022-05-27', 'smithjill@example.com',
  '999913', 'Nicholas', 'Knight', 'Legal', '77612', '2021-09-15', 'matthewbrook@example.com',
  '999914', 'Mindy', 'Myers', 'Finance', '73367', '2022-04-26', 'wilsonwendy@example.com',
  '999918', 'Kimberly', 'Carter', 'Human Resources', '72732', '2023-11-03', 'zc@example.com',
  '999919', 'Jared', 'Christian', 'R&D', '75921', '2023-01-24', 'charles27@example.com',
  '999923', 'Edwin', 'Murphy', 'Sales', '73131', '2022-02-10', 'uortega@example.com',
  '999926', 'Shannon', 'Fry', 'Sales', '70834', '2022-03-02', 'danielpowell@example.com',
  '999929', 'Angela', 'Smith', 'Strategic Planning', '75686', '2021-05-05', 'kg@example.com',
  '999930', 'Douglas', 'Foster', 'Logistics', '74344', '2021-12-03', 'matthew68@example.com',
  '999934', 'Michael', 'Gonzalez', 'Facilities Management', '79035', '2022-10-02', 'm.gonzalez@example.com',
  '999943', 'David', 'Davis', 'Finance', '71909', '2021-03-28', 'steven00@example.com',
  '999944', 'Holly', 'Johnson', 'Product Management', '72765', '2022-08-04', 'r.johnson@example.com',
  '999949', 'Sandra', 'Clay', 'Administration', '71150', '2023-07-29', 'kristen@example.com',
  '999952', 'Nicholas', 'Martinez', 'Strategic Planning', '79877', '2021-02-04', 'n.martinez@example.com',
  '999953', 'Sheila', 'Gutierrez', 'PR', '71869', '2023-01-06', 'nathan55@example.com',
  '999954', 'Robert', 'Dominguez', 'Compliance', '72671', '2021-05-02', 'miles@example.com',
  '999965', 'Anthony', 'Johnson', 'Marketing', '73813', '2022-01-17', 'erin62@example.com',
  '999967', 'Steven', 'Anderson', 'Strategic Planning', '78093', '2021-02-21', 's.anderson@example.com',
  '999969', 'Judith', 'Stewart', 'Facilities Management', '74491', '2023-11-25', 'j.stewart@example.com',
  '999973', 'Natalie', 'Johnson', 'Compliance', '73641', '2022-06-25', 'mgarcia@example.com',
  '999975', 'Samuel', 'Caldwell', 'Logistics', '78381', '2022-11-17', 'slee@example.com',
  '999979', 'Stacey', 'Davis', 'Logistics', '70701', '2021-09-21', 'campbellter@example.com',
  '999980', 'Paige', 'Robertson', 'QA', '78241', '2022-12-28', 'joann79@example.com',
  '999981', 'Veronica', 'Moran', 'QA', '78145', '2022-07-17', 'danielruiz@example.com',
  '999986', 'Kelsey', 'Allen', 'Logistics', '73032', '2023-05-24', 'shannon44@example.com',
  '999989', 'James', 'Campbell', 'Facilities Management', '75228', '2021-05-19', 'j.campbell@example.com',
  '999992', 'Amy', 'Blair', 'R&D', '77185', '2022-02-13', 'jacksonburke@example.com',
  '999993', 'Michael', 'Gutierrez', 'Product Management', '72525', '2021-02-08', 'm.gutierrez@example.com',
  '999995', 'Alexander', 'Nguyen', 'Training and Development', '75383', '2022-01-01', 'a.nguyen@example.com',
  '999997', 'Hannah', 'Hernandez', 'Finance', '73473', '2022-05-22', 'rodriguez@example.com',
  '1000000', 'Tina', 'Price', 'Human Resources', '77657', '2021-09-22', 'johnso@example.com'
]
```